Neural Network, Group 1

In [27]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
%matplotlib inline
```

In [23]:
```
df = pd.read_csv('education_data.csv')
```

TASK 1: Data Set Information

The title of the data set is "Education Inequality Data". It is focused on data that highlights certain aspects of education inequality including the type of school, funding per student, average testing scores, minority percentages, and more.

This data was obtained from -- https://www.kaggle.com/datasets/shamimhasan8/education\-inequality\-data

The purpose of this data is to analyze the different educational experiences that are dependent on many variables and impact unequal learning opportunities.

Dataset Introduction: This dataset contains records for 1000 schools in the United States across 12 states, spanning public, private, and charter institutions at the elementary, middle, and high school levels. Each record captures key resource and demographic indicators, including per-student funding, student-teacher ratio, and internet access rates. The student demographic variables (percent low-income and percent minority) reflect wider socioeconomic diversity across schools. Outcome measures include average test scores and dropout rates. This dataset is well-suited for investigating the relationship between the type of school and the percentage of minority students at that school.

In [8]:
```
education_data_df = pd.read_csv("education_data.csv")
education_data_df.head()
```

Out[8]:

	id	school_name	state	school_type	grade_level	funding_per_student_usd	avg_
0	1	Bowman High School	Michigan	Private	High	9575.80	
1	2	Foster High School	Michigan	Charter	Middle	16733.10	
2	3	Henson High School	New York	Public	Middle	24890.74	
3	4	Weaver High School	Texas	Private	Elementary	6857.49	
4	5	King Elementary School	Michigan	Public	Elementary	16910.73	

TASK 2: Basic Data Exploration

In [10]:
```
education_data_df.head(5)
```

Out[10]:

	id	school_name	state	school_type	grade_level	funding_per_student_usd	avg_
0	1	Bowman High School	Michigan	Private	High	9575.80	
1	2	Foster High School	Michigan	Charter	Middle	16733.10	
2	3	Henson High School	New York	Public	Middle	24890.74	
3	4	Weaver High School	Texas	Private	Elementary	6857.49	
4	5	King Elementary School	Michigan	Public	Elementary	16910.73	

In [11]:
```
education_data_df.tail(5)
```

Out[11]:

	id	school_name	state	school_type	grade_level	funding_per_student_usd	a
995	996	Hale High School	Ohio	Charter	Elementary	8775.63	
996	997	Herrera Elementary School	Georgia	Public	High	24440.93	
997	998	Combs High School	New York	Private	Middle	23947.18	
998	999	Buckley Elementary School	Illinois	Private	Elementary	6459.87	
999	1000	Berg Elementary School	Illinois	Private	Elementary	15471.48	

In [12]:
```
education_data_df.shape
```

Out[12]:
```
(1000, 12)
```

In [13]:
```
education_data_df.columns
```

Out[13]:
```
Index(['id', 'school_name', 'state', 'school_type', 'grade_level',
       'funding_per_student_usd', 'avg_test_score_percent',
       'student_teacher_ratio', 'percent_low_income', 'percent_minority',
       'internet_access_percent', 'dropout_rate_percent'],
      dtype='object')
```

Task 3

Task **3.1:** Use df.info() to display data types of each column.

In [15]:
```
education_data_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 12 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   id                       1000 non-null   int64
 1   school_name              1000 non-null   object
 2   state                    1000 non-null   object
 3   school_type              1000 non-null   object
 4   grade_level              1000 non-null   object
 5   funding_per_student_usd  1000 non-null   float64
 6   avg_test_score_percent   1000 non-null   float64
 7   student_teacher_ratio    1000 non-null   float64
 8   percent_low_income       1000 non-null   float64
 9   percent_minority         1000 non-null   float64
 10  internet_access_percent  1000 non-null   float64
 11  dropout_rate_percent     1000 non-null   float64
dtypes: float64(7), int64(1), object(4)
memory usage: 93.9+ KB
```

Task **3.2:** Identify numerical vs categorical columns.

In [16]:
```
num_cols = education_data_df.select_dtypes(include=['int64', 'float64']).col
cat_cols = education_data_df.select_dtypes(include=['object', 'category', 'b

print("Numerical columns:", list(num_cols))
print("Categorical columns:", list(cat_cols))
```

```
Numerical columns: ['id', 'funding_per_student_usd', 'avg_test_score_percen
t', 'student_teacher_ratio', 'percent_low_income', 'percent_minority', 'inte
rnet_access_percent', 'dropout_rate_percent']
Categorical columns: ['school_name', 'state', 'school_type', 'grade_level']
```

Task **3.3:** Are there any data types that should be converted? Explain why.

No, all the columns containing numerical values have appropriate data types. For example, funding_per_student_usd is currency so it needs to be a float too allow 2 decimal places for cents. Additionallay, the categorical colums have data types of objects which allows the input texts to have things like properties.

In [17]:
```
# TASK: Descriptive Summary
##### Obtain summary statistics.
##### What do these statistics tell you about numerical columns?
##### Are there any unusual or inconsistent values?
##### df.describe() | df.describe(include='all')
```

Task 4

Task **4.1:** Obtain summary statistics.

In [21]:
```
education_data_df.describe(include = 'all')
```

Out[21]:

	id	school_name	state	school_type	grade_level	funding_per_studer
count	1000.000000	1000	1000	1000	1000	1000.0
unique	NaN	695	10	3	3	
top	NaN	Johnson High School	Texas	Private	Elementary	
freq	NaN	12	119	355	345	
mean	500.500000	NaN	NaN	NaN	NaN	15154.1
std	288.819436	NaN	NaN	NaN	NaN	5931.3
min	1.000000	NaN	NaN	NaN	NaN	5012.9
25%	250.750000	NaN	NaN	NaN	NaN	9726.0
50%	500.500000	NaN	NaN	NaN	NaN	15362.4
75%	750.250000	NaN	NaN	NaN	NaN	20280.0
max	1000.000000	NaN	NaN	NaN	NaN	24981.9

Task **4.2:** What do these statistics tell you about numerical columns?

The df.describe(include = 'all') tells us:

There's 1000 values under each of the numerical columns. It shows there are no missing values under any of the numerical columns. It also shows us the mean; standard deviation; minimum and maximum values; the 25%, 50%, and 75% quartiles that show the distribution and median of the values.

Task **4.3:** Are there any unusual or inconsistent values?

In [35]:
```
education_data_df.isna().sum()
```

Out[35]:
```
id                        0
school_name               0
state                     0
school_type               0
grade_level               0
funding_per_student_usd   0
avg_test_score_percent    0
student_teacher_ratio     0
percent_low_income        0
percent_minority          0
internet_access_percent   0
dropout_rate_percent      0
dropout_status            0
dtype: int64
```

In [36]:
```
num_cols = df.select_dtypes(include=['int64', 'float64']).columns
```

```
for col in num_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]
    print(f"{col}: {len(outliers)} outliers")
```

```
id: 0 outliers
funding_per_student_usd: 0 outliers
avg_test_score_percent: 0 outliers
student_teacher_ratio: 0 outliers
percent_low_income: 0 outliers
percent_minority: 0 outliers
internet_access_percent: 0 outliers
dropout_rate_percent: 0 outliers
```

There are no outliers or null values in the colunms. So far there are no indications of an unusual or inconsistent values.

In [20]:
```
# TASK 5: Missing or Duplicate Data
##### Check for missing values in each column.
##### Identify duplicate rows.
##### Provide observations about missing data impact.
##### df.isnull().sum() | df.duplicated().sum()
```

In [24]:
```
education_data_df.isnull().sum
education_data_df.duplicated().sum()
```

Out[24]: 0

In [8]:
```
# TASK 6: Add a New Column
##### Create a new derived column based on existing data.
##### Example: BMI = Weight / Height2
##### Display the first few rows to confirm addition.
##### Briefly explain what this new column represents.
```

In [30]:
```
education_data_df["dropout_status"] = np.select(
    [
        education_data_df["dropout_rate_percent"] > 10,
        education_data_df["dropout_rate_percent"] < 5
    ],
    [
        "High Dropout",
        "Low Dropout"
    ],
    default="Moderate"
)
education_data_df
```

Out[30]:

	id	school_name	state	school_type	grade_level	funding_per_student_usd
0	1	Bowman High School	Michigan	Private	High	9575.80
1	2	Foster High School	Michigan	Charter	Middle	16733.10
2	3	Henson High School	New York	Public	Middle	24890.74
3	4	Weaver High School	Texas	Private	Elementary	6857.49
4	5	King Elementary School	Michigan	Public	Elementary	16910.73
...	...	...	...	...	...	...
995	996	Hale High School	Ohio	Charter	Elementary	8775.63
996	997	Herrera Elementary School	Georgia	Public	High	24440.93
997	998	Combs High School	New York	Private	Middle	23947.18
998	999	Buckley Elementary School	Illinois	Private	Elementary	6459.87
999	1000	Berg Elementary School	Illinois	Private	Elementary	15471.48

1000 rows × 13 columns

In [9]:

```
# TASK 7: Filter the Data
##### Apply at least two different filters to extract subsets.
##### Display the filtered results.
##### Explain what subsets were created and why they are interesting.
##### df[df['column'] > value] | df.query('condition')
```

In [32]:

```
A_score_highschool = education_data_df.query("grade_level == 'High' and avg_

A_score_highschool.head()
```

Out[32]:

	id	school_name	state	school_type	grade_level	funding_per_student_us
67	68	Dean Middle School	Georgia	Charter	High	10475.1
71	72	Mann Elementary School	North Carolina	Public	High	6576.7
79	80	Moore High School	Pennsylvania	Private	High	19502.6
106	107	Allen High School	California	Charter	High	21306.5
114	115	White High School	Michigan	Charter	High	20476.6

In [34]:

```
highfund_highschool = education_data_df.query("grade_level == 'High' and fur
highfund_highschool.head()
```

Out[34]:

	id	school_name	state	school_type	grade_level	funding_per_student_usd
7	8	Boyd High School	Georgia	Private	High	24162.52
12	13	Smith High School	California	Public	High	22343.11
26	27	Johnson High School	Texas	Public	High	21981.75
54	55	Petersen Middle School	New York	Private	High	21626.14
77	78	Garcia High School	Pennsylvania	Private	High	24602.03

Task 7 Explanation

2 Filters

- One filter creates a subset showing only high schools that have a A grade average
- The other filter creates a subset showing only high schools that have 20k or more in funding

Explanation

I just wanted to briefly explore the filtered data and see if there were any quick patterns I could find. I saw from looking at the first 5 values the schools saw schools that had higher funding, so I just looked at the next subset and I saw that there may not be a as strong correlation between funding an average grade percentage as I had thought, just from looking at the first five values of the second filtered subset

In [10]:
```
# TASK 8: Unique Values and Categories
##### List unique values or categories for categorical columns.
##### Identify categorical variables important for analysis or grouping.
##### df[’column’].unique() | df[’column’].value_counts()
```

In [23]:
```
# Categorical columns
categorical_cols = ['state', 'school_type', 'grade_level']

for col in categorical_cols:
    print(f"\nColumn: {col}")
    print("Unique values:", df[col].unique())
    print("\nValue counts:")
    print(df[col].value_counts())
```

```
Column: state
Unique values: ['Michigan' 'New York' 'Texas' 'Pennsylvania' 'Georgia' 'Illi
nois'
 'California' 'North Carolina' 'Ohio' 'Florida']

Value counts:
state
Texas             119
Florida           110
Georgia           107
Illinois          102
Ohio              101
California         97
North Carolina     96
New York           95
Michigan           89
Pennsylvania       84
Name: count, dtype: int64

Column: school_type
Unique values: ['Private' 'Charter' 'Public']

Value counts:
school_type
Private    355
Charter    326
Public     319
Name: count, dtype: int64

Column: grade_level
Unique values: ['High' 'Middle' 'Elementary']

Value counts:
grade_level
Elementary    345
High          330
Middle        325
Name: count, dtype: int64
```

In [11]:

```
# TASK 9: Initial Observations
##### Write a short paragraph summarizing:
##### What you found interesting about the dataset.
##### Any patterns, issues, or questions that emerged during exploration.
##### Think critically about what the data reveals...
```

During our exploration of the dataset, we found it interesting how educational factors vary across different states, school types, and grade levels. States such as Texas and Florida have a higher number of schools represented in the dataset, while others like Pennsylvania and Michigan have fewer. The distribution of school types shows a relatively balanced mix of private, charter, and public schools, which allows for meaningful comparisons between them. We also noticed

that elementary schools appear slightly more frequently than middle and high schools, which may influence overall trends in the data. These patterns suggest that location, school type, and grade level could all play a role in understanding educational inequality. One question that emerged is how these categories relate to outcomes such as funding, test scores, or dropout rates. Overall, the dataset highlights potential disparities in education that warrant further analysis.